

USE CASE: Customer Sales Insights ETL Pipeline

Business Context

A **retail company** stores two datasets in Amazon S3:

1. **Customer data** – contains customer personal details.
2. **Sales transactions** – contains sales made to customers over time.

The company wants to understand **how much each customer has spent** and maintain a **clean dataset** for analytics and reporting.

However:

- The sales data doesn't directly include customer names.
 - The customer table has inconsistent or extra fields not needed for analysis.
 - Both datasets are raw and stored as CSV files in S3.
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Business Goal

Build an **ETL job using AWS Glue Studio (Visual)** to:

- **Extract:** Read both customer and sales datasets from the AWS Glue Data Catalog.
- **Transform:**
 - Select only useful fields (**Sales_Customer_ID**, **Customer_ID**, **First_Name**, **Total_Sales**).
 - Standardize column names and data types.
 - Join both datasets so each sale can be linked to the customer's first name.
 - Aggregate total sales by customer.
- **Load:** Write the cleaned and aggregated result to an S3 location in **CSV format**.

This transformed dataset will then be used for:

- Trend analysis on **high-value customers**.
 - Further data enrichment (e.g., adding location or gender insights).
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End Result

After running the job:

- You'll have a new CSV in S3 showing **each customer's total sales**.
- The business can easily identify top-spending customers.
- The ETL job can be **scheduled or automated** to refresh insights daily. (Improvements)

Step-by-Step

Pre-configuration

In your bucket

`s3://getting-started-with-glue-studio-[UNIQUE-STRING]/datafiles/`

Create another folder `Output/`

Step 1: Open Glue Studio

- Go to **AWS Glue Studio** → **Visual ETL**
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Step 2: Add First Source (Customer Table)

- **Source type:** Click “Data Catalog table”
 - **Database:** `sales_data`
 - **Table:** `customer.csv`
 - It Applies Automatically
 - Rename to “Customer Data Catalog”
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Step 3: Apply Mapping to Customer Data

- While the Customer Data Catalog is highlighted
 - Click the “+” → **Transform** → **ApplyMapping (Change Schema)**
 - Rename to “Customer ApplyMapping”
 - In the mapping:
 - Drop other rows by clicking the checkbox
 - Keep only:
 - `customer_id`
 - `first_name`
 - Rename `customer_id` to a consistent form (`Customer_ID`)
 - Rename `first_name` to a consistent form (`First_Name`)
 - It Applies Automatically
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Step 4: Add Second Source (Sales Table)

- Click **+ Add node** → **Source**
- **Source type:** Data Catalog table
- **Table:** `sales.csv`
- It Applies Automatically

- Rename to “Sales Data Catalog”
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Step 5: Apply Mapping to Sales Data

- While the Customer Data Catalog is highlighted
 - Click the “+” after the Sales source → **Transform** → **ApplyMapping (Change Schema)**
 - Rename to “Customer ApplyMapping”
 - Drop other rows by clicking the checkbox
 - In the mapping:
 - Drop other rows by clicking the checkbox
 - Keep only:
 - `customer_id`
 - `total_sales`
 - Rename `customer_id` → `Sales_Customer_ID`
 - Rename `total_sales` → `Total_Sales`
 - It Applies Automatically
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When applying JOINS, the parents (Sales and Customers) of the Join node should not have overlapping field names.

Step 6: Join Transformed Datasets

- Click **+ Add node** → **Join**
 - In the node parents section
 - Connect the **Customer ApplyMapping** and **Sales ApplyMapping** nodes to the Join.
 - Configure:
 - **Join type:** Inner join
 - **Join conditions** click “Add condition”
 - On the **Customer ApplyMapping** dropdown choose `Customer_ID`
 - On the **Sales ApplyMapping** dropdown choose `Sales_Customer_ID`
 - It Applies Automatically
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Step 7: Aggregate

- While the Join is highlighted
- Click the “+” **Transform** → **Aggregate**
- In settings:
 - **Fields to group by:** `Customer_ID, First_Name`
 - In **Aggregation** click “Aggregate on column”
 - **Field to aggregate:** `Total_Sales`
 - **Aggregate function:** `Sum`
- It Applies Automatically

Step 9: Add Autobalance Processing

- While the Aggregate is highlighted
- Click “+” → **Transform** → **Autobalance Processing**
- **Number of partition: 1**
- **S3 location:**
`s3://getting-started-with-glue-studio-[UNIQUE-STRING]/datafiles/Output/`
- It Applies Automatically

Step 8: Add Target (S3)

- While the Autobalance Processing is highlighted
- Click “+” → **Target**
- **Target type: S3**
- **Format: CSV**
- **Compression Type: None**
- **S3 location:**
`s3://getting-started-with-glue-studio-[UNIQUE-STRING]/datafiles/Output/`
- It Applies Automatically

Step 9: Configure Job Details

- On the tab click on “**Job Details**”
- **Job name:** `customer_sales_etl_job`
- **IAM Role:** Choose your own IAM Role
- **Glue version:** 5.1
- **Worker type:** G.1X
- **Workers:** 2

On the top-right Save and run the job.

Wait a while on the green banner, click “Run details”

Wait for Run status to show **Succeeded**

Now review your s3 Output folder